THE MAXIMAL IDEAL AT A RIG-CLOSED PRIME UNDER RIG-ETALE MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3586-3601, following the statement at 3575-3584.

Editorial note. The final '(details omitted)' is the plug-in check that a finite etale algebra over the local ring A/p is a finite product of fields, and localizing at q keeps exactly one factor; flagged in the notes.

Editorial note. This lemma was consumed as a named input by the carded rig-flat-local-etale (L26273).

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3600 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Etale algebras over a field. Algebra, Lemma algebra-lemma-etale-over-field (cross-chapter, cited at ~3590-3591). B tensor kappa is a finite product of finite separable extensions.
- Finiteness over the complete local base. Algebra, Lemma algebra-lemma-finite-over-complete-ring (cross-chapter, cited at ~3593-3594). B/pB is a finite A/p-algebra.
- Characterization of finite etale maps. Algebra, Lemma algebra-lemma-characterize-etale (cross-chapter, cited at ~3597-3598). Combines finite + flat + etale closed fibre into finite etale.

Source reference. Original reference 00U6: focused-seeds/geometry/S9-restricted-rig-closed-point-etale/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00U3: focused-seeds/geometry/S9-restricted-rig-closed-point-etale/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 031D: focused-seeds/geometry/S9-restricted-rig-closed-point-etale/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$. Assume φ is adic, topologically of finite type, flat, and $A/I \to B/IB$ is étale for some (resp. any) ideal of definition $I \subset A$. Let $\mathfrak{q} \subset B$ be rig-closed such that $\mathfrak{p} = A \cap \mathfrak{q}$ is rig-closed as well. Then $\mathfrak{p}B_{\mathfrak{q}} = \mathfrak{q}B_{\mathfrak{q}}$.*

Proof. Let κ be the residue field of the 1-dimensional complete local ring $A/\mathfrak{p}$. Since $A/I \to B/IB$ is étale, we see that $B \otimes_A \kappa$ is a finite product of finite separable extensions of κ, see Algebra, Lemma 00U3. One of these is the residue field of $B/\mathfrak{q}$. By Algebra, Lemma 031D we see that $B/\mathfrak{p}B$ is a finite $A/\mathfrak{p}$-algebra. It is also flat. Combining the above we see that $A/\mathfrak{p} \to B/\mathfrak{p}B$ is finite étale, see Algebra, Lemma 00U6. Hence $B/\mathfrak{p}B$ is reduced, which implies the statement of the lemma (details omitted). □